

PAPER_873

U_g1=DPM Geophysical Geometry Summation

CP4 Class: #457 Ug1DPMGeophysicalGeometrySummationCalc
Session 200C v5.61 | Source: describe mass without using weight.txt

Abstract

U_g1 is equivalent to the DPM; the total force is a summation over DPM variable forms reflecting unique forces with geophysical geometries. Components: SM_gravity (spherical), U_b buoyancy (spherical, counter-force), resonance (toroidal). Geometry functions G_k enable force diagramming.

Key Equations

```
F_Ug1 = Sum_k [k_k * (fUA1*fSCm1*REB1)*(fUA2*fSCm2*REB2)/r^2 * G_k]
F_SM_gravity = k_grav * DPM1*DPM2/r^2 * sin(theta)
F_Ub = k_buoy * (fUA'*fSCm)/r^2 * sin(theta)
F_resonance = k_res * nu_THz * fSCm * cos(phi)*f(nu_THz)
F_Ug1_total = F_grav - F_buoy + F_resonance
```

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